# Mamoru Ueda

**Robotics & Control Systems Engineer**

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LinkedIn | GitHub | YouTube | 3D Designs

**Visa Status:** Student (20 hours/week) → Graduate Route (Full-time) from June 2026

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## PROFESSIONAL SUMMARY

- **Mechatronic Systems Expert:** PhD candidate with 6+ years of experience designing end-to-end robotic systems, integrating custom PCBs (KiCAD), mechanical structures (Fusion360), and real-time control algorithms (MATLAB/Simulink).
- **Closed-Loop Control Specialist:** Successfully developed and deployed a soft-actuated robotic tail with <40ms latency, utilizing advanced PID control and signal processing pipelines (1.2kHz sample rate).
- **Innovation Driver:** Created the novel "Surprise ERD" neural marker for quantifying tool embodiment, processing complex EEG/EMG data streams to bridge the gap between biological signals and machine response.

## TECH STACK

- **Control & Simulation:** MATLAB, Simulink, PID Control, System Identification.
- **Embedded & Software:** C/C++ (Real-time), Python, Arduino, Raspberry Pi, Jetson Nano, Git.
- **Hardware Engineering:** PCB Design (KiCAD), Circuit Analysis, I2C/SPI/UART, Soldering.
- **Mechanical Design:** Fusion360 (CAD/CAM), 3D Printing (FDM), Mechanism Design.
- **AI & Signal Processing:** DSP, EEG/EMG Analysis, Machine Learning Classifiers.

## EXPERIENCE & PROJECTS

**PhD Researcher: Robotic Tail Embodiment** — *University of Reading, UK*
*Neural & Motor Signatures of Embodiment* — **Sep 2020 – Present**

- **Scope:** Engineered a complete human-in-the-loop robotic system to investigate tool embodiment.
- **Engineering:** Designed a custom PCB and soft-actuator control system (C++) achieving **~40ms latency**, processing multi-modal signals (EEG at **1.2kHz**, EMG at **250Hz**).
- **Algorithm:** Developed a novel "Surprise ERD" signal processing pipeline in MATLAB, utilizing **Time-Frequency Analysis (STFT)** and **Bootstrap Resampling** to quantify prediction errors.
- **Result:** Validated system with **25 participants**; research presented at Eurohaptics 2024 (Top 5 Honourable Mention).

**BCI Systems Engineer: Non-Verbal Communication** — *Mind Storms, UK*
*R&D Project (Part-Time)* — **Oct 2024 – Sep 2025**

- **Scope:** Developed a BCI solution to decode speech intentions for non-verbal users.
- **Engineering:** Implemented a machine learning classifier in MATLAB to translate biological signals into control commands.
- **Result:** Delivered a functional Proof-of-Concept (POC) demonstrated to key stakeholders, securing continued R&D interest.

**Personal Project: DIY Prop Making** — *YouTuber / Maker*
*Iron Man Mask, Psycho-Pass Dominator & More* — **Ongoing**

- **Scope:** Designed and built over 20 unique high-fidelity props using Fusion360 and 3D printing.
- **Engineering:** Integrated custom electronics (Arduino) for synchronized motor control, audio, and LED lighting effects.
- **Result:** Achieved **45k+ views** for top video, showcasing complex electromechanical integration skills.

## EDUCATION

**PhD in Biomedical Engineering** — University of Reading, UK
*Exp. June 2026*

**BSc in Robotics Engineering** — Ritsumeikan University, Japan
*Mar 2019* — (International Competitor: World Robot Summit)

## HONORS & AWARDS

- **Eurohaptics 2024:** Best Work In Progress Honourable Mention (Top 5).
- **SBS Research Conference:** Best Presentation (Top among 20 PhD candidates).
- **JASSO Scholarship:** Type 1 (Top 30%).

## LANGUAGES

**Japanese:** Native **English:** Fluent (Academic & Professional)